Chase Mateusiak

1408 W Washington, Ann Arbor, MI 48103
(989) 859-7075 **|** chase.mateusiak@gmail.com
github.com/cmatKhan **|** orcid.org/0000-0002-2890-4242

WORK

Bioinformatics Scientist *2020–2026*

Washington University, St. Louis

I am a Bioinformatics Scientist for two groups. I have been in the Brent lab since 2020, and joined the Sinha group in January of 2026.

- Michael Brent's lab is in the Center for Genome Sciences & Systems Biology. The Brent lab studies methods of network inference, primarily transcription factor network inference. I perform data management, automate processing workflows, and contribute to the development and distribution of novel network inference methods. My current projects are:
  - Developing a novel method of inferring transcription factor interaction by regressing transcription factor response to either over expression or knockout onto the interaction of multiple TFs. I am the first author on a publication that is current out for review for this project.
  - Developing a harmonized, public data resource of all 11 genome wide transcription factor perturbation response and binding datasets. These data are hosted for free in a computationally efficient format on the machine learning platform HuggingFace.
- Pratik Sinha's lab in the Department of Anesthesiology. The Sinha lab primarily studies clinical data gathered from the ICU. Currently, I am working on the following projects:
  - evaluating the effectiveness of computational cell decomposition (eg CIBERSORTx) for whole blood RNAeq from seriously ill patients.
  - evaluating basis reduction techniques for their utility in analyzing longitudinal RNAseq data in sepsis patient cohorts.
  - refining a method of predicting clock time from bulk RNAseq data in patients who suffer post operative delirium.

Primary responsibilities

- Implement established, recently published, and novel analysis algorithms on bulk RNA-seq, various transcription factor binding modalities (Calling Cards, ChEC-seq, ChIP-seq) and sequence analysis in both single data type and multi-omic modes
- Responsible for data processing and QC of sequencing experiment data for multiple labs, including a large $80 million multi-site collaboration
- Provide computational support to collaborators across organisms (yeast, mouse, human)

Notable achievements

- Developed a novel method of inferring transcription factor interaction at promoters, integrating both transcription factor binding and overexpression and knockout (Mateusiak et al, currently out for review 2026)
- Collected, harmonized, databased all 11 of the major yeast transcription factor binding and perturbation response datasets (Mateusiak et al, expected publication 2026). Notably, the data is hosted using a data lake architecture on huggingface, a machine learning platform that provides free hosting. I developed both a programmatic library and shiny app that provides researchers easy access to this rich data
- Developed, published and released on nf-core a Nextflow workflow for processing Calling Cards data in both yeast and mammals
- Implemented and manage the cloud architecture for two labs, including providing access to Seqera Platform
- Developed independent collaborations resulting in two publications
- Developed the Python curriculum implementation for Algorithms for Computational Biology, a 500-level course in the Washington University CS department
- Trained and directly supervised 5 undergraduate and master's students in machine learning based methods of multi-omics data analysis

Research Assistant *2019*

University of California, San Francisco

- Processed RNA-seq data from human skin cells and tick
- Compiled transcriptome annotations for *Ixodes scapularis*
- Processed data from a deep mutational scan screen in *E. coli*

Teacher *2012–2019*

San Francisco Unified School District

- Responsible for the academic progress of 115 students per year in English and math

Civil Affairs Sergeant *2011–2019*

US Army Reserve

- Trained in developing local relationships, identifying achievable development projects, and coordinating allied forces activities with local government and non-profit organizations

Teacher *2008–2010*

US Peace Corps, Turkmenistan

- Developed and implemented an interactive English as a Foreign Language curriculum that improved speaking and vocabulary skills across eight classrooms for over 250 students

SKILLS

- R/python (advanced)
- Nextflow (advanced)
- Rust
- extensive experience on HPC environments (SLURM, SGE)
- AWS Cloud infrastructure, including terraform
- git/github
- open source best practice packaging in R, Python and Rust with distribution on the major repositories (bioconductor, pypi, bioconda, nf-core)
- containerization in Docker and Singularity
- Structured database development and maintenance in SQL derivatives (postgresql, sqlite) and Django/Django Rest Framework for quickly creating a DB and API
- Data lake architecture and parquet file data storage

EDUCATION

- M.S. Computer Science — Washington University, St. Louis, *2022*
- B.A. English Literature and Language — University of Michigan, Ann Arbor, *2008*

PUBLICATIONS

Mateusiak, C., Jia, E., Plaggenberg, J. N., Erdenebaatar, Z., Wang, Y., Shively, C., Liao, G., Mitra, R. D., & Brent, M. R. (2026). Ubiquitous functional synergy partially explains why most transcription factor binding is non-functional. *bioRxiv* (preprint). https://doi.org/10.64898/2026.01.19.700460

Karagiannis, T. T., **Mateusiak, C.**, Acharya, S., Jung, W. J., Liao, S., et al. (2026). Whole blood transcriptional signatures of age and survival identified in long life family and integrative longevity omics studies. *GeroScience*. https://doi.org/10.1007/s11357-025-02090-x

Abid, D., Brown, H., **Mateusiak, C.**, Doering, T. L., & Brent, M. R. (2025). Mapping the transcriptional regulatory network of a fungal pathogen by exploiting transcription factor perturbation. *mBio*. https://doi.org/10.1128/mbio.02797-25

Kang, Y. S., Jung, J., Brown, H. L., **Mateusiak, C.**, Doering, T. L., & Brent, M. R. (2025). Leveraging a new data resource to define the response of *Cryptococcus neoformans* to environmental signals. *Genetics*, 229(1), 1–29. https://doi.org/10.1093/genetics/iyae178

Yen, A., **Mateusiak, C.**, Sarafinovska, S., Gachechiladze, M. A., Guo, J., Chen, X., Moudgil, A., Cammack, A. J., Hoisington-Lopez, J., Crosby, M., Brent, M. R., Mitra, R. D., & Dougherty, J. D. (2023). Calling Cards: a customizable platform to longitudinally record protein-DNA interactions over time in cells and tissues. *Current Protocols*, 3(9), e883. https://doi.org/10.1002/cpz1.883

Ring, K., Couper, L. I., Sapiro, A. L., Yarza, F., Yang, X. F., Clay, K., **Mateusiak, C.**, Chou, S., & Swei, A. (2022). Host blood meal identity modifies vector gene expression and competency. *Molecular Ecology*, 31(9), 2698–2711. https://doi.org/10.1111/mec.16413

Radkov, A., Sapiro, A. L., Flores, S., Henderson, C., Saunders, H., Kim, R., Massa, S., Thompson, S., **Mateusiak, C.**, Biboy, J., Zhao, Z., Starita, L. M., Hatleberg, W. L., Vollmer, W., Russell, A. B., Simorre, J. P., Anthony-Cahill, S., Brzovic, P., Hayes, B., & Chou, S. (2022). Antibacterial potency of type VI amidase effector toxins is dependent on substrate topology and cellular context. *eLife*, 11, e79796. https://doi.org/10.7554/eLife.79796